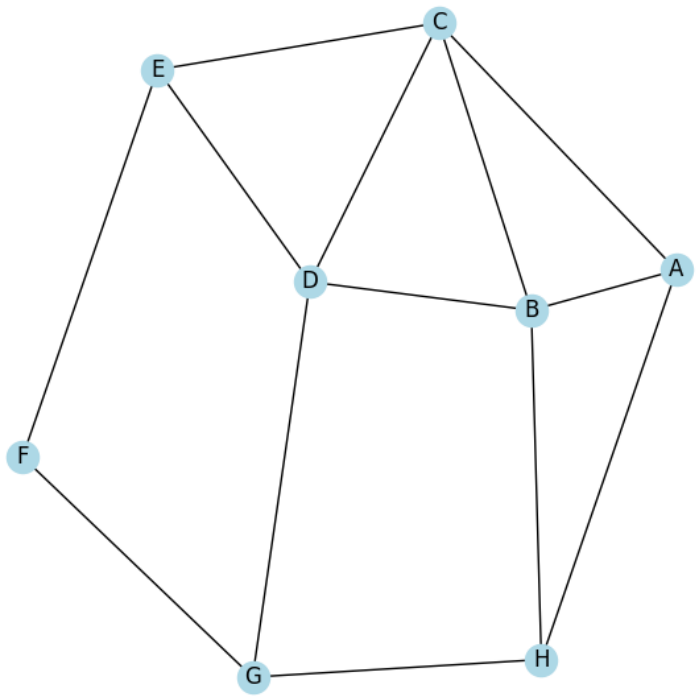
EXPERIMENT NO. 6

INPUT:

```
import pandas as pd
import networkx as nx
import matplotlib.pyplot as plt
import community.community_louvain as community_louvain
df = pd.read_csv(r"C:\Users\ATHARVA\OneDrive\Desktop\social_network.csv")
G = nx.from_pandas_edgelist(df, 'source', 'target')
plt.figure(figsize=(6,6))
nx.draw(G, with_labels=True, node_color='lightblue')
plt.title("Social Network Graph")
plt.show()
partition = community_louvain.best_partition(G)
print("Community Detection Result:")
for node, comm in partition.items():
    print(node, "-> Community", comm)
pagerank = nx.pagerank(G)
print("\nInfluential Users (PageRank):")
for user, score in pagerank.items():
    print(user, ":", round(score,3))
```

social_network.csv:

	A	B
1	source	target
2	A	B
3	A	C
4	B	C
5	B	D
6	C	D
7	C	E
8	D	E
9	E	F
10	F	G
11	G	H
12	H	A
13	D	G
14	B	H

OUTPUT:

```
C:\Users\ATHARVA\OneDrive\Desktop>python sma6.py
Community Detection Result:
A -> Community 0
B -> Community 0
C -> Community 1
D -> Community 1
E -> Community 1
F -> Community 2
G -> Community 2
H -> Community 0

Influential Users (PageRank):
A : 0.114
B : 0.147
C : 0.148
D : 0.149
E : 0.119
F : 0.086
G : 0.12
H : 0.117

C:\Users\ATHARVA\OneDrive\Desktop>
```